

Ashish Jha

Machine Learning Engineer | LLM Training & Evaluations

Göttingen, Germany | work permit available | Ph: +49 155 602 47407

Email | GitHub | LinkedIn | YouTube



Profile

Machine learning engineer with experience building deep learning systems in PyTorch, including large language models, and training them on distributed GPU clusters (Slurm, H100/H200). Experienced in model compression, inference optimization, and training data selection. Author of the open-source training-data-selection tools *GRAFT* and *SAGE*.

Experience

Researcher, Deep Learning

Aug 2025 to Mar 2026

Skolkovo Institute of Science and Technology (Skoltech)

- Researched the Bhattacharyya distance as a training objective across two directions. Primary direction - used it as a classification loss and as a penalty term, training GANs and CNNs in PyTorch and measuring the effect on accuracy and training stability.
- Exploratory direction - studied the Bhattacharyya distance as an alternative to KL divergence in a reinforcement-learning fine-tuning objective for a Mistral LLM.

ML Engineer (MLOps)

Jan 2025 to Mar 2025

Sartorius AG, Göttingen

- Developed end-to-end machine learning pipelines for data preprocessing, feature engineering, model training, and evaluation, enabling the operations team to automate parts of their manual workflow.
- Performed quality assessment and preprocessing on a large industrial dataset (~25,000 records, 200+ features, 20–25% missing values from non-operational sites) and evaluated multiple forecasting models using Azure Databricks.
- Built data ingestion and ETL pipelines in Azure Data Factory, leveraging Apache Spark for scalable data processing.

Researcher, Efficient Deep Learning

Oct 2021 to Dec 2024

Skolkovo Institute of Science and Technology (Skoltech)

- built trained and evaluated both LLMs and CNNs using tools such as PyTorch and tracked experiments in Hugging Face Transformers.
- Ran multi-GPU experiments on Slurm clusters (H100/H200 GPUs), tracked experiments with Weights & Biases.
- Authored papers and open-source software tools that selects representative training subsets; cut training computation by up to 50% at comparable accuracy and published on PyPi.
- Implemented model-compression and inference-optimization methods (structured pruning, INT8/INT4 quantization, knowledge distillation) and compared trade-offs.
- Published in peer-reviewed journals *IEEE Intelligent Systems* and a NeurIPS 2025 workshop.

ML Research Engineer, Model Training & Evaluation

Feb 2022 to Oct 2022

Huawei Technologies & Skoltech (joint research project)

- Designed and trained variational-autoencoder architectures in PyTorch, with customized loss functions to balance between reconstruction quality and bitrate.
- Benchmarked models against strong baselines (JPEG, H.264/H.265) on multiple metrics (PSNR, MS-SSIM, bitrate) across several datasets.

Software Engineer (Python)

Aug 2019 to Jul 2021

Ruijie Networks, St Petersburg

- Developed Python tools and internal CLI libraries for data collection, log parsing and anomaly detection across network infrastructure.
- Built data extraction and processing pipelines (REST APIs, JSON, CSV) to analyse high-volume system data.

- Contributed to setting up automated testing and validation workflows, cutting manual testing effort by around 40%.

Technical Executive, Data & Infrastructure

Aug 2016 to Feb 2019

CG Communications Ltd., Nepal

- Maintained cloud infrastructure for a payment platform and automated deployment workflows utilizing Bash and Python for automation scripts, roughly improving uptime from 80 to around 97%.

Education

M.Sc. Computer Science

2019 to 2021

Saint Petersburg Electrotechnical University (LETI)

B.E. Electronics & Communication Engineering

2010 to 2014

Visvesvaraya Technological University (VTU)

Publications

- *Lightweight Attribute Localizing Models for Pedestrian Attribute Recognition*. IEEE Intelligent Systems, 2026.
- *SAGE: Streaming, Agreement-driven Gradient Sketches for Representative Subset Selection*. Workshop paper, NeurIPS 2025.

Software

- *GRAFT*, an open-source PyTorch library for training-data selection, released on PyPI.
- *SAGE*, streaming, agreement-driven gradient sketches for representative subset selection (NeurIPS 2025 workshop).
- *LW-ALM*, lightweight attribute-specific localization model for pedestrian attribute recognition (IEEE Intelligent Systems).

Skills

Training at scale	PyTorch, Hugging Face Transformers; multi-GPU / distributed training on Slurm (H100/H200); Weights & Biases;
Data & pipelines	ETL (Apache Spark, Azure Data Factory); Pandas, NumPy, SQL.
Evaluation & optimization	model compression and inference optimization (structured pruning, INT8/INT4 quantisation, knowledge distillation).
Model architectures	Transformers, convolutional networks, (variational) autoencoders; classification, attribute recognition and localization; learned image and video compression; image-quality metrics (PSNR, MS-SSIM).
Infrastructure	Docker, FastAPI, REST APIs, Git, GitHub Actions, Azure ML, GCP.
Programming	Python, Bash; C++ (basic).
Languages	English (fluent), German (A2), Hindi (native), Nepali (native), Russian (limited working).